Lecture-16
Conservation Laws for Control Volumes IV: Conservation of Angular Momentum

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Learning Objectives

By the end of this lecture, you should be able to

1. Define **torque and angular momentum** in the context of fluids
2. Understand how Reynolds Transport Theorem (RTT) extends **Newton's laws of motion to control volumes**
3. Differentiate situations requiring the use of the **linear momentum equation versus the angular momentum equation**

Angular momentum principle

- **All torques acting on a control volume** result in a **net rate of change of angular momentum within the control volume**, and a **net flux of angular momentum across the control surfaces**.

$$\vec{r} \times \vec{F}_s + \int_{CV} \vec{r} \times \vec{g}\, \rho\, d\forall + \vec{T}_{shaft} = \frac{\partial}{\partial t} \int_{CV} \vec{r} \times \vec{V}\, \rho\, d\forall + \int_{CS} \vec{r} \times \vec{V}\, \rho\, \vec{V} \cdot d\vec{A}$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

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